Q.26 What is a DC load line? What is its importance?

Q.27 Write the comparison between BJT and JFET.

Q.28 Differentiate between N-Type and P-Type extrinsic semiconductor.

Q.29 Why capacitors are connected in parallel and inductor in series in filter circuits?

Q.30 Explain Schottky diode.

Q.31 Explain the energy band structure of Si & Ge.

Q.32 Draw and explain the atomic structure of P-Type semiconductor.

Q.33 Discuss the working of single stage amplifiers in CE configuration.

Q.34 Explain the working of half wave rectifier with suitable waveforms.

Q.35 Draw the characterisitics of zener diode and explain it.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Compare CB, CC and CE configurations.

Q.37 Draw the circuit diagram of full wave bridge rectifier and explain its working along with waveforms.

Q.38 Differentiate semiconductor, conductor and insulator on the basis of energy band structure. Why semiconductors have negative temperature coefficient of resistance?

No. of Printed Pages : 4 120933/030933

Roll No. ....

3rd Sem / Electrical Engg., GE, Power Station Engg., Elect. & Eltx. Engg., Fire Tech & Safety

Sub : Electronics – I/Basic Electronics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Diffusion capacitance depends on

a) Area of junction

b) Permittivity

c) Width of depletion region

d) Mean lifetime of carriers

Q.2 Which of the following is not a semiconductor?

a) Lead b) GaAs

c) Silicon d) Germanium

Q.3 The value of forbidden energy gap in insulators is of the order of

a) 0 eV b) 1 ev

c) 5 ev d) Less than 2 ev

Q.4 The biasing circuit which gives best stability to Q-point is

a) Emitter resistor biasing

b) Potential dividerbiasing

c) Feedback resistor biasing

d) Base resistor biasing

Q.5 The functions of emitter in NPN transistor is

a) To emit or inject holes into collector
b) To emit or inject electrons into collector
c) To emit or inject electrons into base
d) To emit or inject holes into base

Q.6 A field effect transistor is essentially a

a) Current driven device
b) Voltage driven device
c) Power driven device
d) None of the above

Q.7 In case of ideal current sources, they have

a) Large value of current
b) Low value of voltage
c) Zero internal resistance
d) Infinite internal resistance

Q.8 What is Zener diode used as?

a) Regulator   b) Oscillator
c) Filter   d) Rectifier

Q.9 JFET is a _____ controlled device

a) Resistance   b) Proton
c) Current   d) Voltage

Q.10 The purpose of coupling capacitor in an amplifier is to

a) Control frequency
b) Match impedance
c) Limit bandwidth
d) Prevent DC mixing with output

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What is the purpose of transistor in electronic circuits?

Q.12 Capacitance can be defined as the ratio of electric charge on each conductor to what between them?

Q.13 What are multistage amplifiers?

Q.14 What is JFET?

Q.15 Draw PNP transistor.

Q.16 Give graphical representation of current source

Q.17 The number of diodes required in a bridge rectifier circuit is/are _____.

Q.18 What is the value of ripple factor of full wave rectifier?

Q.19 What type of device is a diode?

Q.20 Why the stabilization of operating point is needed?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Deduce the relation between a, b and g.

Q.22 What are filter circuits? Explain any one filter circuit.

Q.23 What ish-parameters? Why are they used?

Q.24 Give the characteristics of a MOSFET.

Q.25 Draw the frequency response of single stage transistor amplifier.